# Tactix proof repair and readiness checkpoint

Codex for Andrew

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**Outcome.** The AM442 Claude Code campaign, run explicitly with Opus, finished its online-configuration affected scope successfully across 750,000 candidate executions. It also exposed a pre-existing native-LRAT proof defect in ordinary solving. Tactix Sweep had reused a derivation prefix whose hints had been normalized for a different unit target. Commit `421a38a1` retains the raw prefix and normalizes it separately for each target. The old exact proof rejects under CakeLPR; the patched reduced 28-clause and original 129-clause H1 proofs both return `s VERIFIED UNSAT`. The focused Sweep test and the complete Sweep DRAT/LRAT/native-DSR proof suite pass. This is a scoped repair, not combined N2 acceptance. The AM444 Opus owner is preparing its affected fuzz; full Debug job 1723644 is queued on the committed source.

**Canonical H2 proof.** The public X7 action remains legal. The canonical route uses no-option `dsr-trim`, then LSR and Trestle. On the reduced 16-clause H2 input, moving one deletion before the first derived unit produces a proof verified by both checker directions and Trestle. On the original 42-clause replay, moving six retained-reason deletions ahead of the first unit yields a backward elaboration independently verified by Trestle, but forward checking rejects the first unit: those clauses also support its derivation. Neither checker-only edit is a general emitter repair. AM443's two-variable negative control shows a reason can already be implied by the original input, leaving no earlier proof-line window. The diagnostic implied unit deletion option changes which proofs verify in both directions, so it has not been adopted as a checker contract.

**Reference rule coverage.** The AM446 source crosswalk maps the nonincremental CaDiCaL proof producers and SC24 Kissat/AE-Kissat-MAB clausal producer families to Tactix's typed actions. Their emitter interfaces expose RUP, RAT, deletion and finite substitution witnesses; this static mapping does not prove public reachability or correct current certificates. Satsuma's VeriPB mode and selected LymphoSAT winner paths use general pseudo-Boolean arithmetic, redundancy/dominance and subproofs. Tactix has terminal startup PBP for Lucky 19/20 but no general live PB action or proof interface. Particular winner formulas have independently verified Tactix clausal/startup proofs; the general interface gap remains a deliberate architecture question.

**Gates and learning readiness.** The paired AES diagnostic is terminal: both original-CNF native DSR proofs verify after 5,780/5,790 checker seconds. Its frozen matched Tier-C pair is still unscorable at the 4,800-second checker limit. The older full Debug job 1723391 had passed 186/187 tests, with its final scripted demo running at this checkpoint. AM427 Sweep lifecycle Opus arrays are live. No current combined Rank 1–50, B28+C5, WV100, WV2-89, N4 reference-parity or Tier-D result follows from these scoped checks. The existing online configuration supports prototyping dose and order, but production learning still needs a shared adjustable inprocessing controller and

owned live switches for branching and clause deletion. The final readiness verdict must follow the correctness, fuzz and performance gates.

**Next actions.** Collect the two Debug suites and both Opus owners, repair any bounded handoff, and solve H2 without suppressing legal public X7 actions. Then check one combined pin against independent proofs/models, the required random-policy and default-policy fuzzing, and the full performance and reference-parity gates. Only after N1–N4 and the final learning-readiness review pass should the 1,166-instance Tier D comparison launch at 5,000 seconds, four CPUs and 32 GiB per instance.

**Evidence.** `benchmarks/reports/2026-09-25-am444-sweep-lrat-core-target-repair.md`; `benchmarks/reports/2026-09-25-am444-h2-early-deletion-control.md`; `benchmarks/reports/2026-09-25-am445-h2-original-deletion-order-control.md`; `benchmarks/reports/2026-09-25-am446-reference-proof-producer-crosswalk.md`; the AM442 and AM427 owner status/report files under `benchmarks/runs/`.